UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 24347

LH EVOLUTION OF COSMIC STRUCTURE

SEMESTER 2 EXAMINATIONS 2022/23

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

A list of useful formulae is provided at the end of the paper.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

Two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. (a) Consider a matter-only universe where the mass energy-density parameter Ω_m = 1 and the Hubble parameter is H. Starting from the equation for the evolution of linear perturbations in an expanding universe,

$$\ddot{\delta} = \frac{3}{2}\Omega_m H^2 \delta - 2H\dot{\delta},$$

calculate how the overdensity, δ, depends on time, t. **[6]**

(b) In the spherical top hat model of overdensities, a system has a maximum radius at turnaround of $R_{\rm max}$. The time at which the system reaches its minimum radius occurs at time $t_{\rm vir}$ = $\pi\alpha R_{\rm max}$, where α is a constant related to the energy of the system. If this system collapses from turnaround and virialises, calculate its final virialised density relative to the critical density of the universe ($\rho_{\rm crit}$). Assume Ω_m = 1.

[Hint: recall that the evolution of the radius of a region in the absence of a perturbation is $R_{\rm un} = \frac{R_{\rm max}}{4}\left(\frac{12t}{\alpha R_{\rm max}}\right)^{2/3}$] **[8]**

(c) Consider a universe in which the dark matter particle has been discovered and has a mass of 3.1 keV. This particle becomes non-relativistic during the radiation dominated epoch and the present day cosmic microwave background (CMB) temperature is T_0 = 2.7 K. On what comoving length scale would you expect to see an effect on the growth of structure from the dark matter particle?

[In this universe, in the radiation dominated epoch, the cosmic time, t, is

$$t = 6 \times 10^4 \text{ yr} \left[\frac{10^4 \text{ K}}{T}\right]^2$$

where T is the CMB temperature.] **[8]**

(d) Describe the Sunyaev-Zeldovich (S-Z) effect observed in galaxy clusters, and show quantitatively that the surface brightness of the S-Z effect is independent of redshift. If you have an independent measurement of the redshift to the galaxy cluster, explain how a measurement of the S-Z effect could be used to measure the Hubble parameter H. **[8]**

2. (a) Describe the key stages in the evolution of the baryonic component of our Universe from the epoch at which the energy density of matter and radiation were equal, until the current epoch. **[8]**

(b) Consider a flat universe with the present-day Hubble parameter H_0 = 70 km s^{-1} Mpc^{-1} and the present-day mass energy-density parameter $\Omega_{\mathrm{M},0}$ = 0.3. In the Press-Schechter formalism the comoving density, $\frac{dn}{d\log M}$, of halos of mass M is

$$\frac{dn}{d\log M} = \sqrt{\frac{2}{\pi}}\frac{\rho_{\mathrm{M}}}{M}\left|\frac{n_{\mathrm{eff}}+3}{6}\right|\nu e^{-\nu^2/2},$$

where ν = $1.69/\sigma_{\mathrm{M}}$, $\sigma_{\mathrm{M}} \propto M^{-(n_{\mathrm{eff}}+3)/6}$, n_{eff} is the effective spectral index, and ρ_{M} is the comoving matter density.
Calculate the present-day comoving density of halos with mass of 5×10^{14} M$_\odot$ in a universe where the normalization of the power spectrum on scales of 8 Mpc σ_8 = 0.79 and n_{eff} = -1.53. **[8]**

(c) In a hypothetical flat, matter-only universe (Ω_M = 1) that contains only hydrogen, the electron scattering depth from the cosmic microwave background is found to be τ_e = 0.09. Determine the redshift of reionization in this hypothetical universe if the present day Hubble parameter is H_0 = 70 km s^{-1} Mpc^{-1} and the present day electron density is n_e = 5.0 $\times$ 10^{-7} cm^{-3}. Assume that reionization is abrupt.
[Hint: The optical depth is τ_e = $\int n_e \sigma_T dx$, where σ_T is the Thomson cross section (6.65 $\times 10^{-25}$ cm^2) and x is the proper length.] **[8]**

(d) Explain why the sub-millimetre region of the electromagnetic spectrum is especially useful in the search for high redshift star-forming galaxies. Could this method be used to find the very first galaxies? Why or why not? **[6]**

3. (a) The gas fraction of a galaxy cluster $f_{\rm gas}$ has been found to scale with the gas temperature, T, as $f_{\rm gas} \propto$ T$^{0.25}$. Assuming spherical symmetry, derive an expression describing how X-ray luminosity L_x depends on gas temperature T. **[10]**

(b) Consider a singular isothermal dark matter halo characterized by its circular velocity V_c which is independent of radius. The density profile of the halo is $\rho(r)$ =$V_c^2/(4\pi G r^2)$, where r is the distance from the centre of the halo.

i. Find an expression for the total mass M_{200} of the halo as a function of V_C and the Hubble parameter H. **[8]**

ii. The spin parameter, λ, of a rotating system can be defined as $\lambda = \frac{J/M}{RV}$, where J is the angular momentum, M is the mass, R is the radius and V is the velocity. Assume that a forming disk galaxy has the same specific angular momentum as the dark matter halo within which it is embedded.
Find an expression for the radius of the disk galaxy in terms of the dark matter halo radius (R_{200}) and dark matter spin parameter (λ_{DM}).
[Note: Recall that the angular momentum of a disk embedded in a halo is given by $J_d = 2M_d R_d V_c$, where M_d is the disk mass and R_d is the disk radius.] **[5]**

(c) Consider a galaxy with a stellar mass of 7 $\times$ 10^{10} M$_\odot$ which contains a black hole at its centre with a mass 1 $\times$ 10^8 M$_\odot$. The stellar initial mass function of this galaxy results in one supernovae produced for every 100 M$_\odot$ of stars formed, and each supernovae explodes with a total energy of 10^{51} ergs. Does the supernova feedback or the AGN feedback have a stronger effect on this galaxy? Quantitatively support your answer and state any assumptions you have made. **[7]**

Formula Sheet

Friedmann equation
$H^2 = \frac{8}{3}\pi G\rho - \frac{k}{a^2}$

Fluid equation
$\dot{\rho} + 3\frac{\dot{a}}{a}\left(\rho + P\right) = 0$

Acceleration equation
$\frac{\ddot{a}}{a} = -\frac{4}{3}\pi G\left(\rho + 3P\right)$

Equation of State
$P = \omega\rho$

Critical density
$\rho_c = \frac{3H^2}{8\pi G}$

Hubble parameter
$H = \frac{\dot{a}}{a}$

$H(z) = H_0\sqrt{\frac{\Omega_{M,0}}{a^3} + \frac{\Omega_{r,0}}{a^4} + \Omega_{\lambda,0}}$

Scale factor (Matter only)
$a(t) \propto t^{2/3}$

Scale factor (Radiation only)
$a(t) \propto t^{1/2}$

Overdensity
$\delta(r,t) = \frac{\rho(r,t) - \bar{\rho}(t)}{\bar{\rho}(t)}$

Dynamical time (gravitational)
$t_{dyn} = \frac{1}{\sqrt{4\pi G\bar{\rho}}}$

Press-Schechter number density
$\frac{dn}{d\log M} = \sqrt{\frac{2}{\pi}}\frac{\rho_{\rm M}}{M}\left|\frac{n_{\rm eff}+3}{6}\right|\nu e^{-\nu^2/2}$

NFW density profile
$\rho(r) = \frac{4\rho_s}{(r/r_s)(1+r/r_s)^2}$

Optical depth
$\tau_e = \int n_e \sigma_T dx$

Feedback energy from supernova
$E_{fb} = \epsilon_{SN}\,\xi\,M_*\,E_{SN}$

Feedback energy from black holes
$E_{AGN} = \epsilon_{BH}\,M_{BH}\,c^2$

Virial theorem
$\mathrm{PE} + 2\,\mathrm{KE} = 0$

Radiation pressure
$P_{rad} = \frac{L}{c}\frac{1}{4\pi R^2}$

Ram pressure condition
$\rho_{icm}v^2 = 2\pi G\Sigma(R)\Sigma_g(R)$

X-ray luminosity
$L_x \propto \rho^2\,V\Lambda(T)$

Correlation function
$\langle N_g\rangle = n[1 + \xi(r)]V$

Integrated flux density of the SZ effect
$S_\nu = j(x)\int d\Omega \int \frac{k_B T_e}{m_e c^2}\sigma_T n_e dl$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^{3} W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Solar radius	$R_\odot$	6.95×10^{8} m
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

This page is intentionally blank.

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.